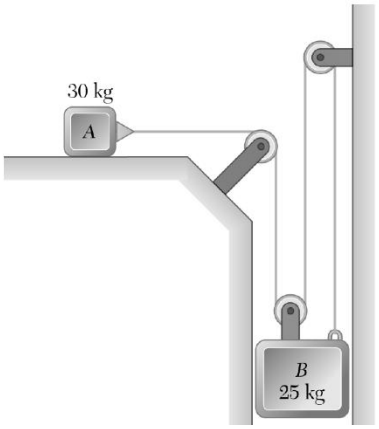
동역학 숙제 #3
(김병일 교수)

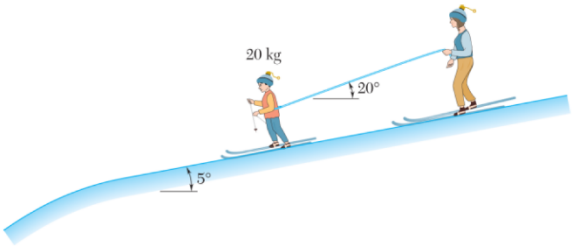
(Due in Two Weeks)

문제 1)



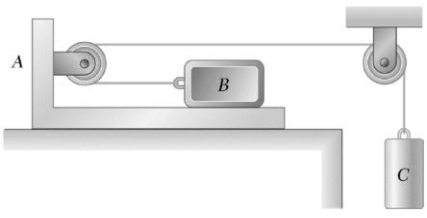
The two blocks shown are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and between block A and the horizontal surface, determine (a) the acceleration of each block, (b) the tension in the cable.

문제 2)



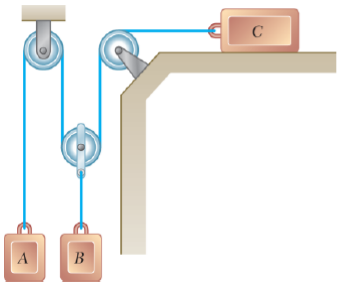
A mother and her child are skiing together, and the mother is holding the end of a rope tied to the child's waist. They are moving at a speed of 7.2 km/h on a gently sloping portion of the ski slope when the mother observes that they are approaching a steep descent. She pulls on the rope with an average force of 7 N. Knowing the coefficient of friction between the child and the ground is 0.1 and the angle of the rope does not change, determine (a)the time required for the child's speed to be cut in half, (b)the distance traveled in this time.

문제 3)



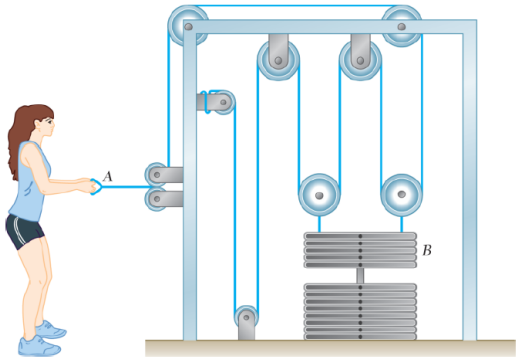
A 6-kg block B rests as shown on a 10-kg bracket A. The coefficients of friction are $\mu_s = 0.30$ and $\mu_k = 0.25$ between block B and bracket A, and there is no friction in the pulley or between the bracket and the horizontal surface. (a) Determine the maximum mass of block C if block B is not to slide on bracket A. (b) If the mass of block C is 10% larger than the answer found in a, determine the accelerations of A, B and C.

문제 4)



Knowing that μ = 0.30, determine the acceleration of each block when $m_A = m_B = m_C$.

문제 5)

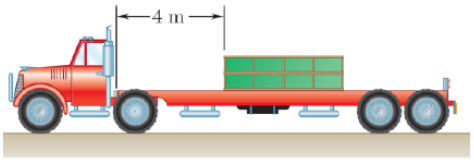


An athlete pulls handle A to the left with a constant force of P = 100 N. Knowing that after the handle A has been pulled 30 cm its velocity is 3 m/s, determine the mass of the weight stack B.

문제 6)

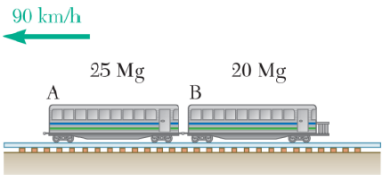
An airplane has a mass of 25 Mg and its engines develop a total thrust of 40 kN during take-off. If the drag **D** exerted on the plane has a magnitude $D = 2.25v^2$, where v is expressed in meters per second and D in newtons, and if the plane becomes airborne at a speed of 240 km/h, determine the length of runway required for the plane to take off.

문제 7)



The coefficients of friction between the load and the flat-bed trailer shown are $\mu_s = 0.40$ and $\mu_k = 0.30$. Knowing that the speed of the rig is 72 km/h, determine the shortest distance in which the rig can be brought to a stop if the load is not to shift.

문제 8)



A light train made up of two cars is traveling at 90 km/h when the brakes are applied to both cars. Knowing that car A has a mass of 25 Mg and car B a mass of 20 Mg, and that the braking force is 30 kN on each car, determine (a) the distance traveled by the train before it comes to a stop,(b) the force in the coupling between the cars while the train is showing down.